

Paper I — The Certified Budget

What Authenticity Costs in Metres: A Certified Budget for Cryptography and Sensing in UAV Navigation

Integrated proposal, merging P1 and P2 · RobustUAVs.ai · Roger Nick Anaedevha · 2026-10-01
Target: IEEE Symposium on Security and Privacy, main track · follow-on to the parent manuscript [1]

Abstract

A robustness certificate for UAV navigation bounds the position error an adversary can induce as the product of a delay it can hide, a rate at which that delay becomes position error, and a dynamics factor. We observe that both of the first two factors are security-engineering choices, and that a single operation — authentication — moves them in opposite directions. Authenticating a navigation source lets the estimator trust it, which lowers the error rate γ ; authenticating a message costs signing, verification and transmission time, which raises the delay Δ . The certificate is therefore a budget in which cryptography and sensing trade, and it decides the net sign. We derive the crossover condition under which authentication increases the certified mission floor, show that tightening an intrusion detector makes authentication *more affordable but less beneficial*, and predict that the binding cost of post-quantum signatures on a flight-controller bus is occupancy, a cliff, rather than per-message latency, a slope. We propose to measure γ for three navigation sources and the authentication cost of five schemes on flight-controller-class hardware, to map the crossover across transports, and to show that an adversary who floods a bus with invalid signatures turns the defence into the delay attack the certificate already bounds.

1 The thesis

The parent manuscript [1] certifies a mission-completion floor by bounding the worst-case position error

$$\rho(\theta) = \Delta(\theta) \cdot \gamma \cdot e^{LT_c} \leq m, \quad (1)$$

where $\Delta(\theta) \leq H \min(\theta, s)$ is the delay an adversary can hide below a detector set at θ across H hops, γ is metres of error per second of degraded input, L is the local Lipschitz constant of the navigation dynamics, T_c the certification horizon, and m the corridor margin. Two of the four inputs are fixed by physics (L) and regulation (m). The other two are not.

- γ depends on *which navigation source the estimator trusts*. P1 observed that it is a property of the source, not of the attack, and that an authenticated source can be trusted under conditions where an unauthenticated one cannot.
- Δ is indifferent to *why* an input is stale. P2 observed that authentication itself makes inputs stale, by exactly the quantity the theorem already bounds.

Read together: **authentication is both a credit and a debit against one certified budget**. It buys a lower γ and pays for it in Δ . Neither proposal alone could ask which effect wins. This paper asks it, answers it with a closed-form condition, and measures where real transports and schemes fall.

2 Threat model: what authentication actually buys

We inherit the parent’s adversary — a compromised mesh relay that forwards every message late, and a GNSS jammer or spoofer — and must be precise about what authentication does against each, because the obvious answer is wrong.

Against forgery, authentication works. A spoofed navigation fix that fails verification is rejected rather than fused. This is P1’s mechanism, and it is what lowers γ : the estimator stops absorbing the spoofer’s position offset.

Against delay, authentication alone does nothing. A validly signed message forwarded four seconds late is still validly signed. The parent’s time-delay adversary is untouched by signatures. Authentication bounds delay only when the signature covers a timestamp and the receiver rejects messages older than a freshness window τ_{fresh} . MAVLink 2 signing already carries such a timestamp [11]. Freshness then caps hidden delay at $\tau_{\text{fresh}} + \epsilon_{\text{clk}}$, where ϵ_{clk} is the receiver’s clock error.

The clock couples back to the attack. On a UAV the reference clock is very often disciplined by GNSS. Under jamming it free-runs, and ϵ_{clk} grows with holdover duration. So the freshness bound weakens precisely when the navigation attack is active. This coupling is, to our knowledge, unmodelled in both the drone-authentication and the certified-robustness literature, and a security reviewer will raise it first. The paper treats it as a first-class term:

$$\Delta_{\text{eff}}(\theta, t) = \min(H \min(\theta, s), \tau_{\text{fresh}} + \epsilon_{\text{clk}}(t)) + \Delta_{\text{auth}}. \quad (2)$$

Symmetric MACs do not stop a compromised relay. MAVLink 2’s HMAC-SHA256 uses a shared key. A compromised relay holds that key and can sign anything. Against the parent’s adversary, therefore, only asymmetric signatures or delayed-disclosure schemes such as TESLA [8] provide source authentication — and those are exactly the expensive ones. This is why the cost side of the budget matters and is not a footnote.

New adversary: induced verification. An adversary on the bus sends frames carrying invalid signatures. The receiver must spend verification time before it learns they are invalid, so the attacker converts the defence’s own cost into delay (§7).

3 The certified budget

Let S be the set of available navigation sources with authentication states a , and κ an authentication scheme operating at message rate r on a transport with per-frame payload b and frame duration t_f . The merged budget is

$$\rho(\theta, S, a, \kappa) = \Delta_{\text{eff}}(\theta) \cdot \gamma_{\text{eff}}(S, a) \cdot e^{LT_c} \leq m. \quad (3)$$

Write $E = e^{LT_c}$, and in the delay regime, where re-anchoring holds by construction and the freshness cap is not binding, $\Delta_{\text{eff}} = \Delta(\theta) + \Delta_{\text{auth}}$.

Proposition 1 (Crossover). *Let authenticating the navigation path change the effective rate from γ_u to $\gamma_a < \gamma_u$ at cost Δ_{auth} . Authentication strictly reduces the certified error ρ if and only if*

$$\Delta_{\text{auth}} < \Delta(\theta) \left(\frac{\gamma_u}{\gamma_a} - 1 \right). \quad (4)$$

Proof. $(\Delta + \Delta_{\text{auth}}) \gamma_a E < \Delta \gamma_u E$, divide by $\gamma_a E > 0$ and rearrange. $\square$

The condition is elementary; what is not elementary is that its two sides are measured on entirely different instruments — a logic analyser on a CAN bus and a flight log of a jammed receiver — and that nobody has put them in one inequality.

Proposition 2 (More affordable, less beneficial). *Let $B(\theta) = m/(\gamma_a E) - \Delta(\theta)$ be the absolute authentication budget and $C(\theta) = \Delta(\theta)(\gamma_u/\gamma_a - 1)$ the crossover threshold of Eq. (4). For $\theta < s$, B is decreasing in θ and C is increasing in θ . Tightening the detector therefore leaves more room to afford authentication and less reason to want it.*

Proof. $\Delta(\theta) = H\theta$ on $\theta < s$; B has slope $-H$ and C has slope $H(\gamma_u/\gamma_a - 1) > 0$. $\square$

This refines P2’s first proposition, which stated only the affordability half. The two halves answer different questions — *can I pay for it?* and *is it worth paying for?* — and an engineer who conflates them will over-invest in cryptography on a platform whose detector is already tight.

Proposition 3 (The cost is a cliff, not a slope). Δ_{auth} decomposes into compute, transmission and protocol terms,

$$\Delta_{\text{auth}} = \underbrace{t_{\text{sign}} + t_{\text{verify}}}_{\text{compute}} + \underbrace{\lceil \sigma/b \rceil t_f \cdot q(U)}_{\text{transmission}} + \underbrace{t_{\text{proto}}}_{\text{protocol}},$$

where σ is the tag size and $q(U)$ a queueing factor in bus occupancy $U = U_0 + r \lceil \sigma/b \rceil t_f$. Since $q(U) \rightarrow \infty$ as $U \rightarrow 1$, every scheme has a maximum admissible message rate $r^* = (1 - U_0) / (\lceil \sigma/b \rceil t_f)$, above which Δ_{auth} is unbounded and the certificate is vacuous regardless of m . Ordering schemes by σ captures only the transmission term: schemes that minimise it, such as TESLA, can maximise t_{proto} through their key-disclosure delay.

This is P2’s third proposition made precise, and it is the one we expect to carry the empirical weight (§4).

Proposition 4 (Two regimes). *Under the time-delay adversary, re-anchoring holds by construction and the crossover of Proposition 1 decides. Under a persistent-denial adversary that withholds every unauthenticated source for longer than T_c , the unauthenticated certificate is vacuous, and authentication of at least one source is necessary for any certified floor.*

This is P1’s second claim, stated as a regime boundary rather than a universal.

The selection rule. Given m , θ , a transport, and a cost per source and scheme, Eq. (3) induces a feasibility set over (suite, scheme) pairs; the rule returns its cost-minimal element. This is P1’s third claim, extended to choose the scheme as well as the suite. It is verified against brute-force enumeration; the optimality argument goes to the appendix.

4 What the parent’s constants already say

Two things can be computed today, before any new measurement, from constants the parent manuscript already reports: $H = 2$, $\theta = 0.25$ s, $L = 1.181$, $T_c = 1$ s, and the campaign supremum $\gamma = 1.625$ m/s. They reproduce the parent’s $\gamma_{\text{req}} = 6.14$ m/s at $m = 10$ m, which is the check that the arithmetic below is wired to the same numbers.

m (m)	window edge θ^* (s)	certified at $\theta = 0.25$ s?	authentication budget B (s)
2	0.189	no	−0.122
5	0.472	yes	0.445
10	0.945	yes	1.389
20	1.889	yes	3.278

[DERIVED] from measured parent constants; B uses $\gamma_a = \gamma_u = 1.625$ m/s, i.e. assumes authentication buys no γ improvement.

Two readings follow. First, at $m = 2$ m nothing certifies *even at zero authentication cost*: P2’s second proposition is already partly a consequence of the parent’s results, and the new content is naming that margin per transport. Second, at $m = 5$ m the budget is 0.445 s. Per-message signing latency on a microcontroller is measured in milliseconds, so **per-message latency is very unlikely to be the binding constraint**. Something else must be.

Order-of-magnitude motivation for the cliff. The table below is a back-of-envelope calculation, not a result, and is labelled as such. Tag sizes are from the standards [7]; the frame model assumes classic CAN at 1 Mbit/s, seven payload bytes per DroneCAN multi-frame transfer frame, and roughly 130 bits per frame including stuffing.

Scheme	tag (B)	frames	ms / message	bus saturates at (msg/s)
HMAC-SHA256 (MAVLink 2 signature field)	13	2	0.26	~ 3800
Ed25519	64	10	1.3	~ 770
ML-DSA-44 (FIPS 204)	2420	346	45	~ 22
SLH-DSA-SHA2-128s (FIPS 205)	7856	1123	146	~ 7

[ESTIMATE] — to be replaced by measurement (§6).

If these estimates are even roughly right, an ML-DSA-44 signature costs about 45 ms per message — comfortably inside a 445 ms budget — yet saturates a classic CAN bus at about 22 signed messages per second, below the update rate of many attitude and inertial streams. That is Proposition 3 in numbers: the scheme is affordable per message and inadmissible per second.

■ *These are the most quotable numbers in the proposal and the least established. They rest on an assumed frame duration and an assumed per-frame payload, and real buses carry a base load U_0 that only lowers the saturation rates further. They motivate the measurement; they must never appear in the paper except as the prediction the measurement tested.*

5 Research questions

RQ1 How much does γ depend on the navigation source? Measured for GNSS (done: 1.625 m/s), inertial-only, and visual-inertial.

RQ2 What does authentication cost on real flight hardware? Compute, transmission and protocol terms of Δ_{auth} , for five schemes, two targets, three transports.

RQ3 Where is the crossover? For each (transport, scheme, source) triple, the sign of Eq. (4) and the margin at which it flips.

RQ4 Does the freshness bound survive GNSS denial? The growth of ϵ_{clk} under holdover, and the denial duration after which Eq. (2) reverts to the detector bound.

RQ5 How much delay can an attacker induce through verification? And how much of it does a pre-filter recover?

6 Measurement plan

6.1 γ per navigation source (RQ1)

1. **Inertial-only — first, and this week.** From the Whelan flight logs already on disk [12]: freeze the GNSS fix at onset and integrate the inertial measurements. This replaces an error-budget *assertion* in the parent’s cruise discussion with a *measurement*, so it strengthens the parent paper as well as this one.
2. **Visual-inertial.** Public VIO datasets with UAV motion profiles, using the parent’s baseline-corrected estimator γ_δ rather than the cumulative one, which the parent showed is contaminated by detector threshold.
3. **LEO signals of opportunity — appendix only.** No receiver logs are available. A `physics_model` value derived from published RMSE [5, 4] is reported, tagged, and kept out of every headline number.

6.2 Δ_{auth} per scheme, target and transport (RQ2)

- **Schemes:** HMAC-SHA256 (MAVLink 2 baseline, shared key), Ed25519, ML-DSA-44, SLH-DSA-SHA2-128s [7], and TESLA-style delayed-key disclosure [8], whose t_{proto} is an explicit tunable parameter and so gives the cleanest test of Proposition 3.

- **Targets:** an STM32H7-class microcontroller (as on Pixhawk 6) and a companion-computer-class ARM board. Cycle-accurate timing on the first, wall-clock under load on the second.
- **Transports:** classic CAN (DroneCAN), CAN-FD, and MAVLink 2 over a serial telemetry link.
- **Validation:** the fragmentation model is checked against inter-frame timing of the benign portion of the HCRL UAVCAN capture [6], which also supplies a realistic base load U_0 .
- **Amortisation as a design axis:** per-message signing, batch signing over k messages, and MAC-plus-periodic-signature hybrids each trade bandwidth for latency, which is precisely the budget's trade.

6.3 Clock holdover (RQ4)

Determine first whether the released flight logs carry both a GNSS-disciplined time and a free-running system time. If they do, $\epsilon_{\text{clk}}(t)$ under the recorded jamming intervals is directly measurable. If they do not, use oscillator holdover specifications for the flight-controller crystal, tagged `physics_model`.

■ *We do not yet know whether the Whelan logs carry both clocks. This must be checked before RQ4 is promised, and if they do not, RQ4 becomes a modelled analysis and the paper says so.*

7 The induced-verification attack (RQ5)

An adversary with bus access injects frames carrying syntactically valid but cryptographically invalid signatures at rate r_{adv} . Each costs the receiver t_{verify} before rejection, and on a multi-frame scheme it also costs reassembly and bus occupancy. The induced delay on legitimate navigation traffic enters Eq. (2) exactly as adversarial staleness does, so the composition theorem bounds the mission-level effect.

Positioning against prior art. Signature-flooding denial of service is not new: avoiding per-packet signature verification was a design motivation for TESLA [8]. The contribution is not the mechanism. It is (i) the *mission-level bound* on what the attack achieves, through the certificate, and (ii) the observation that post-quantum migration makes it markedly worse on constrained buses, because lattice verification is costlier than the classical verification it replaces and multi-frame reassembly must complete before verification can begin.

Mitigation in the same section. Cheap pre-filters (a truncated MAC checked before the full signature), per-source rate limits, and verification budgets, each evaluated by how much of the induced Δ it recovers. As in the parent's evasion analysis, the output is a defensive configuration procedure, and the attack is presented alongside it rather than ahead of it.

8 Contributions, as they would appear in the paper

- **A certified budget** in which sensing and cryptography trade through a single robustness certificate, with a closed-form crossover for when authentication helps (Prop. 1), the affordability–benefit inversion (Prop. 2), and a two-regime boundary (Prop. 4).
- **The freshness–clock coupling:** an explicit model of how GNSS denial weakens the freshness check that is authentication's only defence against delay.
- **Measurements** of γ for three navigation sources and of Δ_{auth} for five schemes on flight-controller-class hardware across three transports, every number provenance-tagged.
- **The cliff:** evidence that bus occupancy, not per-message latency, determines which post-quantum schemes a UAV can adopt (Prop. 3).
- **An attack with its bound and its mitigation:** induced verification, bounded at the mission level.

- **A selection rule** returning the cheapest (sensor suite, scheme) pair that keeps a given detector setting certified.

9 Relation to prior work

Line of work	What it reports	What this paper adds
Post-quantum authentication for drones [10, 9]	signature size, cycles, energy	what those milliseconds do to the aircraft
Broadcast authentication (TESLA) [8]	cheap source authentication at the price of disclosure delay	that delay priced in metres, against a certified margin
Alternative PNT [5, 4]	positioning accuracy under denial	accuracy translated into a certified window
Certified robustness [2, 3]	stability under given perturbations	where the perturbation comes from, and what defending against it costs
The parent manuscript [1]	a certificate with γ and Δ as fixed inputs	both as design variables, jointly optimised

10 Relation to the parent manuscript’s limitations

Limitation	Closed?	How
The certificate assumes re-anchoring	fully	Prop. 4 makes it a regime boundary with a named satisfying mechanism.
γ from one platform in one regime	partly	Adds sources; does <i>not</i> deliver a cruise campaign, which stays data-gated.
One attack surface	partly	Adds the bus and the C2 link on the defence side.
Evidence-class imbalance	no	Unchanged.

11 Risks and fallbacks

- **The crossover could be one-sided.** If every measured Δ_{auth} is far below $C(\theta)$, authentication always wins and Proposition 1 has no interesting boundary. The cliff makes this unlikely on classic CAN, but if it happens the paper reports it plainly: “authentication always pays, except where occupancy forbids it” is still a result, and the occupancy boundary becomes the headline.
- **Hardware access.** A development board is inexpensive; a full flight controller is needed only for bus validation. Fallback: a pinned ARM target, tagged as such. The crossover survives because it depends on the ratio $\Delta_{\text{auth}}/\Delta(\theta)$, which a pinned target still bounds.
- **Only two γ sources measured.** GNSS and inertial-only already establish that γ is source-dependent. VIO strengthens; it is not load-bearing.
- **The freshness coupling might dominate.** If ϵ_{clk} grows fast enough under holdover, freshness provides no protection within the certification horizon and the paper’s message becomes “authenticate the source, not the timing”. That is a valid and useful outcome.
- **Scheme selection dates.** Pin to FIPS 203/204/205 and publish the model, so a new scheme can be placed without re-running the campaign.

12 Page plan and timeline

Section	pp	Section	pp
Introduction	1.25	Δ_{auth} results and the cliff	1.5
Background and threat model	1.25	Crossover map and selection	1.25
The certified budget	2.0	Induced verification	1.25
Measurement methodology	1.5	Discussion and limitations	0.75
γ results	1.25	Related work, conclusion	1.0
Body total		13.0	

About five months. Weeks 1–2: inertial γ and the budget calculator (both need nothing new). Weeks 3–10: hardware benchmarks, VIO, fragmentation validation. Weeks 8–12: the attack and its mitigation. Weeks 13–20: writing, with Paper II submitted first so that its vocabulary is citable.

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